APPARATUS GROUPS & TEMPERATURE CLASSES FOR COMMON FLAMMABLE MATERIALS

Gas/Vapour	Temperature class	Apparatus group
Acetic acid	T1	IIA
Acetone	T1	IIA
Acetylene	T2	IIC
Ammonia	T1	IIA
Butane	T2	IIA
Cyclohexane	T3	IIA
Di-ethyl ether	T4	IIB
Ethanol (ethyl alcohol)	T2	IIA
Ethylene	T2	IIB
Gasoline (petrol)	T3	IIA
Hydrogen	T1	IIC
Kerosene	T3	IIA
Methane (natural gas) (non-mining)	T1	IIA
Methanol (methyl alcohol)	T2	IIA
Methyl ethyl ketone (MEK)	T2	IIB
Propane	T1	IIA
Propan-1-ol (n-propyl alcohol)	T2	IIB
Propan-2-ol (iso-propyl alcohol)	T2	IIA
Toluene	T1	IIA
Xylene	T1	IIA

Dust Apparatus Groups

IIIA	Combustive flyings
IIIB	Non-conductive dust
IIIC	Conductive dust

Dusts	Typical Ignition Temperature (°C) Cloud	Layer
Aluminium	590	>450
Coal dust (lignite)	380	225
Flour	490	340
Grain dust	510	300
Methyl cellulose	420	320
Phenolic resin	530	>450
Polythene	420	(melts)
PVC	700	>450
Soot	810	570
Starch	460	435
Sugar	490	460

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TEMPERATURE CLASS (GROUP II)

Maximum surface temperature	T Class
450°C	T1
300°C	T2
200°C	T3
135°C	T4
100°C	T5
85°C	T6

N.B. For Group I applications, apparatus has rigid 150°C (coal dust) and 450°C (methane) limits rather than T classes.

SIRA ATEX CERTIFICATE NUMBER

SIRA 09 ATEX 1234

Serial number

Reference to ATEX 95 directive

Year of certification

Name of notified body performing EC-type examination

Suffixes:
U – component certification X – special conditions for safe use apply

TYPICAL EQUIPMENT MARKING (ELECTRICAL EQUIPMENT)

IEC Ex

Ex Supplies Ltd
Chester CH4 9JN
United Kingdom
CE 0518
Floodlight P7
Ex II 2 G D
Ex de IIB T4 Gb
Ex tb IIIC T135°C Db IP66
Ta = -20°C to +60°C
SIRA 09ATEX1234
IECEx SIR 09.0788
2009 Sn 0001

- IECEx licenced mark (IECEx only)
- Name and address of manufacturer
- CE mark and number of notified body responsible for quality audit (ATEX only - CAT 1 & 2 electrical, CAT 1 non-electrical)
- Series or type designation
- Specific marking of explosion protection (ATEX only)
- Equipment Group and Category
- Type of explosive atmosphere G - gases, vapours or mists, D - dust
- Apparatus Group and T Class
- Equipment Protection Level (EPL)
- Protection Concept(s)
- Ambient Temperature (-20°C to +40°C if not marked)
- Certificate number (ATEX only)
- Certificate number (IECEx only)
- Year of manufacture and serial number

TYPICAL EQUIPMENT MARKING (NON-ELECTRICAL EQUIPMENT)

CONTENTS OF AN ATEX TECHNICAL FILE (FOR CATEGORY 2/3 NON-ELECTRICAL EQUIPMENT)

Ref: Annex VIII (Internal Control of Production)

- General description of product
- Design/manufacturing drawings
- List of standards applied
- Applicable test reports
- Ignition hazard assessment

Ex Supplies Ltd, Chester CH4 9JN
Scirocco Mk II Pump
30/600 r.p.m IP65
File ref. no. 108112
Serial number: 1234
Manufactured: 2009
CE Ex II 2 G D c k T4

- Technical file reference
- Equipment group/category & type of explosive atmosphere
- Protection Concept(s)
- T Class

COMPLIANCE ROUTES AND EQUIPMENT SELECTION

Zone Gas	Zone Dust	Equipment Category	Relevant ATEX Annexes for Compliance	Group	Hazardous Area Characteristics
0	20	1	III and IV or V	II	Explosive atmosphere present continuously or long periods or frequently (>1000 hours/year)
1	21	2*	III and VII or VI	II	Explosive atmosphere is likely to occur in normal operation occasionally (>10 <1000 hours/year)
1	21	2**	VIII#	II	
2	22	3	VIII	II	Explosive atmosphere is not likely to occur in normal operation or infrequently and for short periods (<10 hours/year)
Mining		M1	III and IV or V	I	If explosive atmosphere present, equipment remains energised
Mining		M2*	III and VII or VI	I	If explosive atmosphere present, equipment de-energised
Mining		M2**	VIII#	I	If explosive atmosphere present, equipment de-energised
Any		Any	IX≠	Any	–

* Electrical equipment and internal combustion engines only

** Non-electrical equipment only

\# and communicate the technical file to a notified body

≠ Alternative route for any product

NORTH AMERICAN APPROVAL MARKING

NORTH AMERICAN / EUROPEAN AREA CLASSIFICATION EQUIVALENTS - GAS

	Flammable Material Present Continuously	Flammable Material Present Intermittently	Flammable Material Present Abnormally
IEC / EU	Zone 0	Zone 1	Zone 2
US NEC 505	Zone 0	Zone 1	Zone 2
US NEC 500	Division 1		Division 2
CA CEC Section 18	Zone 0	Zone 1	Zone 2
CEC Annex	Division 1		Division 2

IEC classification per IEC 60079-10 IEC classification per EN 60079-10
US classification per ANSI / NFPA 70 National Electrical Code® (NEC®) Article 505
CA classification per CSA C22.1 Canadian Electrical Code (CEC) Section 18 or Annex J

NORTH AMERICAN / EUROPEAN EQUIPMENT GROUPING EQUIVALENTS - GAS

Typical gas	US (NEC® 505) CA (CEC Section 18) EU IEC	US (NEC® 500) CA (CEC Annex J)
Acetylene	Group IIC	Class I / Group A
Hydrogen	(Group IIB + H_2)	Class I / Group B
Ethylene	Group IIB	Class I / Group C
Propane	Group IIA	Class I / Group D
Methane	Group I*	Mining*

** Not within scope of NEC®. Under jurisdiction of MSHA. Not within scope of CEC.*

NORTH AMERICAN / EUROPEAN AREA CLASSIFICATION EQUIVALENTS - DUST

	Combustible Dust Present Continuously	Combustible Dust Present Intermittently	Combustible Dust Present Abnormally
IEC / EU	Zone 20	Zone 21	Zone 22
US NEC 506	Zone 20	Zone 21	Zone 22
US NEC 500	Division 1		Division 2
CA CEC Section 18	Division 1		Division 2

US area classification per ANSI / NFPA 70 National Electrical Code® (NEC®) Article 500 or 506
CA area classification per CSA C22.1 Canadian Electrical Code (CEC) Section 18
EU area classification per EN 61241-10 IEC area classification per IEC 61241-10

NORTH AMERICAN / EUROPEAN EQUIPMENT GROUPING EQUIVALENTS - DUST

Typical material	EU (60079-0) IEC (60079-0)	US (NEC 506)	IEC (61241-0) EU (61241-0)	US (NEC 500) CA (CEC Section 18)
Metal dusts	IIIC	N/A	D	Class II, Group E
Carbonaceous dusts	IIIB	D	D	Class II, Group F
Non-conductive dusts	IIIB	D	D	Class II, Group G
Fibres and flyings	IIIA	D	D	Class III

ATEX 100a
Directive 94/9/EC

ATEX 137
99/92/EC & DSEAR

sira CERTIFICATION

Sira Test & Certification Ltd Rake Lane, Eccleston, Chester, CH4 9JN, England

Tel: +44 (0) 1244 670 900 Fax: +44 (0) 1244 681 330 Email: info@siracertification.com Website: www.siracertification.com

PROTECTION CONCEPTS

Electrical	Symbol	Typical IEC EPL	Typical Zone(s)	IEC Standard (status at August 2009)	Basic concept of protection
Optical Radiation	Op pr Op sh Op is	Ga Gb Gc	0,1,2 0,1,2 0,1,2	IEC 60079-28 IEC 60079-28 IEC 60079-28	Inherently safe protected. Protected by shutdown
Increased safety Type 'n' (non-sparking)	e nA	Gb Gc	1,2 2	IEC 60079-7 IEC 60079-15	No arcs, sparks or hot surfaces. Enclosure IP54 or better
Flameproof Type 'n' (enclosed break)	d nC	Gb Gc	1,2 2	IEC 60079-1 IEC 60079-15	Contain the explosion, quench the flame
Quartz/sand filled	q	Gb	1,2	IEC 60079-5	Quench the flame
Intrinsic safety Intrinsic safety Intrinsic safety Type 'n' (energy limitation)	ia ib ic nL	Ga Gb Gc Gc	0,1,2 1,2 2 2	IEC 60079-11 IEC 60079-11 IEC 60079-11 IEC 60079-15	Limit the energy of sparks and surface temperatures
Pressurised (up to 2007) Pressurised Pressurised Pressurised Type 'n' (sealing & hermetic sealing) Type 'n' (restricted breathing) Type 'n' (simple pressurised) Encapsulation Encapsulation Oil immersion	p px py pz nC nR nZ ma mb o	Gb Gb Gb Gc Gc Gc Gc Ga Gb Gb	1,2 1,2 1,2 2 2 2 2 0,1,2 1,2 1,2	IEC 60079-2 IEC 60079-2 IEC 60079-2 IEC 60079-2 IEC 60079-15 IEC 60079-15 IEC 60079-15 IEC 60079-18 IEC 60079-18 IEC 60079-6	Keep the flammable gas out

Dust Protection (Electrical)	Symbol	Typical IEC EPL	Typical Zone(s)	IEC Standard (status at August 2009)	Basic concept of protection
Enclosure	t	Da Db Dc	20, 21, 22	IEC 61241-1	Standard protection for dusts, rugged tight enclosure
Intrinsic safety	i	Da Db Dc	21, 22	IEC 61241-11	Similar to t, but with some relaxations if circuit inside is intrinsically safe
Encapsulation	m	Da Db Dc	22	IEC 61241-18	Protection by encapsulation of incendive parts
Pressurised	p	Db Dc	21, 22 22	IEC 61241-2	Protection by pressurisation of enclosure

Non-electrical	Symbol	Typical IEC EPL	Possible Zone(s)	EN Standard (status at August 2009)	Basic concept of protection
General	-	-	0,1,2	EN 13463-1	Low potential energy
Flow restricted enclosure Flameproof enclosure	fr d	- -	2 1,2	EN 13463-2 EN 13463-3	Relies on tight seals, closely matched joints and tough enclosures to restrict the breathing of the enclosure
Constructional safety	c	-	0,1,2	EN 13463-5	Ignition hazards eliminated by good engineering methods
Control of ignition sources	b	-	0,1,2	EN 13463-6	Control equipment fitted to detect malfunctions
Pressurisation	p	-	1,2	EN 13463-7	Enclosure is purged and pressurised to prevent ignition sources from arising
Liquid immersion	k	-	0,1,2	EN 13463-8	Enclosure uses liquid to prevent contact with explosive atmosphere

INGRESS PROTECTION (IP) CODE BS EN 60529 (IEC 60529)

FIRST NUMERAL	SECOND NUMERAL
Protection against solid objects	**Protection against water**
0 - No special protection	0 - No special protection
1 - Objects > 50 mm diameter (e.g. part of a hand)	1 - Vertically dripping water
	2 - Vertically dripping water, when enclosure tilted by 15°
2 - Objects > 12.5 mm diameter (e.g. finger)	3 - Sprayed water up to 60° from the vertical
3 - Objects > 2.5 mm diameter (e.g. tool)	4 - Sprayed water from all directions
	5 - Water jets
4 - Objects > 1.00 mm diameter (e.g. wire)	6 - Powerful water jets
	7 - Temporary submersion to a depth of 1m
5 - Dust protected	
6 - Dust tight	8 - Extended submersion to a depth > 1m

NEMA TESTING APPROXIMATE EQUIVALENT TO IPXX

NEMA TYPE	IP
NEMA 1	IP10
NEMA 2	IP11
NEMA 3	IP54
NEMA 3R	IP14
NEMA 3S	IP54
NEMA 4 and 4X	IP55
NEMA 5	IP52
NEMA 6 and 6P	IP67
NEMA 12 and 12K	IP52
NEMA 13	IP54

This table cannot be used to convert IP codes to NEMA ratings

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- **Quality Assurance & Repair**
- **ATEX 137 & DSEAR**
- **IEC 61508 Functional Safety**
- **Training and Competence**
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UKAS TESTING 0327 UKAS PRODUCT CERTIFICATION 011 IECEx IEC sira CompEx

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Assessment

Knowledge Certificate
If sufficient and relevant

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